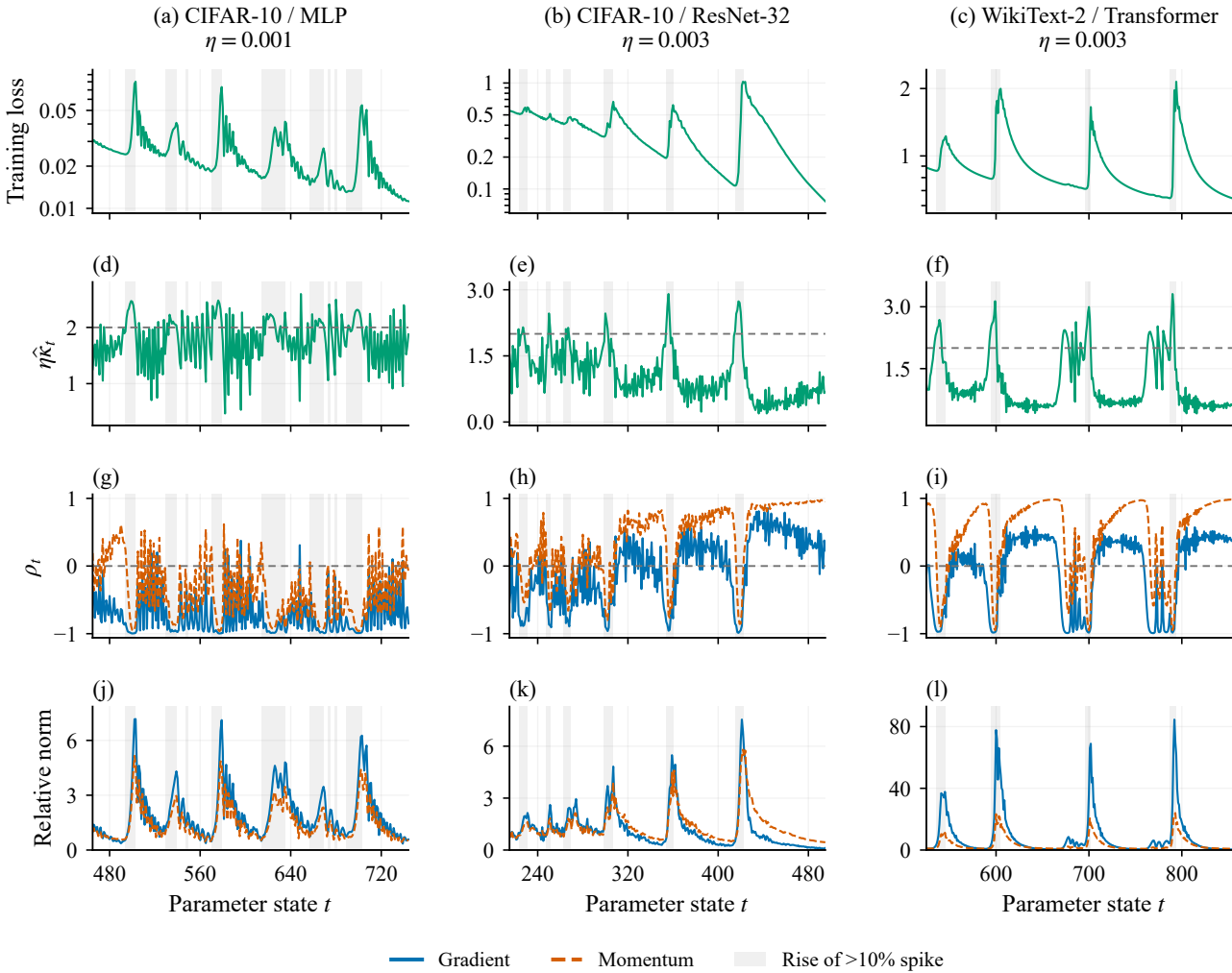

(a) CIFAR-10 / MLP
$\eta = 0.001$
(b) CIFAR-10 / ResNet-32
$\eta = 0.003$
(c) WikiText-2 / Transformer
$\eta = 0.003$
Training loss
(d)
(e)
(f)
$\eta\hat{\kappa}_t$
(g)
(h)
(i)
ρ_t
(j)
(k)
(l)
Relative norm
Parameter state t
Parameter state t
Parameter state t
Gradient
Momentum
Rise of >10% spike